



Tests of pin-ended cold-formed lean duplex stainless steel columns



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ABSTRACT

This paper describes a test program on cold-formed lean duplex stainless steel columns compressed between pinned ends. Two square hollow sections and four rectangular hollow sections were tested at different column lengths. The material properties of the test specimens were obtained from tensile coupon tests and stub column tests. The test specimens were cold-rolled from flat strips of lean duplex stainless steel (EN1.4162). The column specimens were concentrically loaded between pinned ends. The ultimate loads and the failure modes of each column are presented. The American, Australian/New Zealand and European specifications for stainless steel structures are assessed by comparing the column test strengths and available data in the literatures with the design strengths. It should be noted that these specifications do not cover the material of lean duplex stainless steel. A reliability analysis was carried out to assess the current design rules of stainless steel for lean duplex material. Generally, the specifications are able to predict the strengths of the tested columns. The design approach of using full cross-section area and material properties obtained by stub column tests for all classes of sections including slender sections was recommended. This recommended design approach does not require section classification and calculation of effective area, and provides a more accurate and less scattered prediction than those using the current design rules.

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1. Introduction

Cold-formed stainless steel sections have been increasingly used in architectural and structural applications, due to their aesthetic appearance, superior corrosion resistance, ease of maintenance, and ease of construction. Yet the high price of stainless steel limits the application to construction projects. Currently, most of the stainless steel structures are using austenitic and ferritic grades. The lean duplex stainless steel, a relatively new material, offers an opportunity for stainless steels to be used more widely in structural engineering due to the more competitive cost, increased price stability combined with good mechanical properties and appropriate levels of corrosion resistance [1]. The lean duplex stainless steel of grade EN 1.4162 (LDX2101) in this study contains around 1.5% nickel composition only, compared with more than 5% in most of the common stainless steel grades. Nevertheless, little research has been found on the lean duplex material, and it is not covered in current design specifications. There is an eminent need to recommend design rules for lean duplex stainless steel structural members to facilitate its wider application.

Extensive research was performed on austenitic stainless steel grades EN 1.4301 (AISI 304) and EN 1.4401 (AISI 316) as well as duplex stainless steel grade EN 1.4462 (S31803) structural members. The behavior of duplex and high strength austenitic stainless steel sections was investigated by Young and Lui [2]. Gardner and Nethercot [3]

investigated the stress–strain curves and cross-sectional data of hollow sections grade EN 1.4301 stainless steel. The material properties of lean duplex stainless steel were investigated by Huang and Young [4], including residual stresses, tensile coupon tests, and stub column tests. Theofanous and Gardner [5] carried out experimental and numerical investigations on stub columns and pin-ended columns of lean duplex stainless steel square hollow section (SHS) and rectangular hollow section (RHS) columns. It is suggested that the class 3 limit (c/t) be relaxed from 30.7 to 37.0, and a modified effective width equation has been proposed. Theofanous and Gardner [6] carried out three-point bending tests to investigate the structural behavior of lean duplex stainless steel and proposed design rules. Fixed-ended stainless steel columns were tested by Liu and Young [7], and it was concluded that the American Specification (ASCE), Australian/New Zealand Standard (AS/NZS) and European Code (EC3) were generally conservative except for some of the long columns. Pin-ended column tests and beam tests were carried out by Gardner and Nethercot [8] to investigate the structural behavior of cold-formed austenitic stainless steel hollow sections, and it was found that the EC3 is overly conservative. Rasmussen and Rondal [9] investigated the application of column curves in EC3 for stainless steel alloys, and proposed column curves for austenitic, ferritic and duplex alloys. They also mentioned that there is a need for further testing of duplex and ferritic stainless steel columns. Gardner [10] investigated a continuous method of cross-section classification and member design. The member strengths obtained from the proposed design method were compared with test strengths and design strengths predicted using EC3. It was shown that the proposed design method

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offers an average increase in member capacity and a reduction in scatter of the prediction. Up to date, little research is being carried out on structural behavior of lean duplex stainless steel. Therefore, it is necessary to investigate the suitability of the current design specifications for lean duplex stainless steel columns design.

Pin-ended axial compression tests on cold-formed lean duplex stainless steel columns were performed in this study. Four RHS and two SHS with different column lengths were tested. Initial overall geometric imperfections of the column specimens were measured prior to testing. The column specimens were concentrically loaded between pinned ends. The test strengths of the lean duplex column specimens obtained from in this study and previous researches [4,5] were compared with the design strengths predicted by the American Specification [11], Australian/New Zealand Standard [12] and European Code [13] for stainless steel structures, aiming at assessing the applicability of these design specifications to lean duplex stainless steel columns. Reliability analysis was carried out for the cold-formed lean duplex stainless steel columns. Design approach using stub column properties and full area for all classes of sections including slender sections are recommended in this study. Characterized by simpler calculation procedure and utilization of the full cross-sectional material properties in all classes of sections, the recommended design strengths are shown to provide a generally more accurate and less scattered predictions than the current design rules for the specimens in this study.

2. Test specimens

Pin-ended axial compression tests were performed on cold-formed lean duplex stainless steel (EN 1.4162) columns, which is a relatively new and high strength material. The test specimens were cold-rolled from flat strips. The test program consists of six series including two SHS (series C2 and C3) and four RHS (series C1, C4, C5 and C6). The nominal section sizes of series C1 to C6 are $50 \times 30 \times 2.5$, $50 \times 50 \times 1.5$, $50 \times 50 \times 2.5$, $70 \times 50 \times 2.5$, $100 \times 50 \times 2.5$, and $150 \times 50 \times 2.5$, respectively. The specimens of lengths ranging from 200 mm to 1550 mm were tested between pinned ends by aiming at zero eccentricity for each column. Therefore, the specimens were considered as concentrically loaded compression members. The length of the stub column was equal to three times the depth of the cross-section dimension. The stub column tests have been reported in Huang and Young [4]. Tables 1–6 show the measured cross-section dimensions of each test specimen using the nomenclature defined in Fig. 1. The cross-section dimensions as shown in Tables 1–6 were obtained by the average measured values at both ends of each specimen. Both ends of the specimens were milled flat and then welded to 20 mm thick steel end plates for the specimens to be connected to the end bearings. The longest column lengths were designed to produce l_e/r_y ratio of 140, 83, 86, 81, 79 and 76 for series C1 to C6, respectively, so as to obtain a column curve for each series. The test specimens were labeled such that the test series and the specimen length could be identified. For example, the label C5L900R defines the

Table 2
Measured specimen dimensions for series C2.

Specimen	Depth	Width	Thickness	Outer radius	Inner radius	Area
	<i>D</i> (mm)	<i>B</i> (mm)	<i>t</i> (mm)	<i>r_o</i> (mm)	<i>r_i</i> (mm)	<i>A</i> (mm ²)
SC2L150 ^a	50.5	50.5	1.493	3.0	1.0	285.4
C2L200	50.2	50.3	1.527	2.8	1.2	290.2
C2L550	50.6	50.6	1.493	3.0	1.0	286.0
C2L900	50.5	50.5	1.507	3.2	1.2	288.1
C2L1200	50.5	50.4	1.497	2.7	1.0	285.9
C2L1550	50.8	50.7	1.510	2.8	1.0	290.2
Mean	50.5	50.5	1.507	2.9	1.1	288.3
COV	0.005	0.004	0.009	0.060	0.082	0.008

^a Reported in Huang and Young [4].

following specimen. The first two letters indicate that the column specimen belongs to test series C5. The third letter “L” indicates the length of the specimen. The last three or four digits are the nominal length of the specimen in millimeters (900 mm). If a test was repeated, the last letter “R” indicates the repeated test.

3. Material properties and initial geometric imperfections

Tensile coupon tests of flat portions of cross-section were conducted for each test series to determine the material properties of the test specimens, as detailed in Huang and Young [4]. The untested specimens where the tensile coupons were extracted are of the same batch as the pin-ended column specimens. The material properties obtained from the coupon tests, including the static 0.2% proof stress ($\sigma_{0.2}$), static tensile strength (σ_u), initial Young's modulus (E_o), elongation at fracture (ϵ_f), and Ramberg–Osgood parameter (n), are summarized in Table 7. The parameter n is to describe the shape of the initial non-linear part of the stress–strain curve, and was obtained from the measured 0.01% ($\sigma_{0.01}$) and 0.2% ($\sigma_{0.2}$) proof stresses using $n = \ln(0.01/0.2) / \ln(\sigma_{0.01}/\sigma_{0.2})$. The complete and the initial part of the static stress–strain curves obtained from the tensile coupon test for series C4 are shown in Figs. 2 and 3, respectively.

Stub column tests were also carried out to determine the material properties of the test specimens for the complete cross-section in the cold-worked state [4]. The material properties obtained by stub column tests include the effect of residual stress and the strength enhancement in the section due to the cold-forming process. Table 8 shows the measured material properties obtained from the stub column tests, which include the static 0.2% proof stress ($\sigma_{0.2}$), static tensile strength (σ_u), initial Young's modulus (E_o), parameter n that calculated using the same method as the tensile coupon tests, and failure mode, where “L” indicates local buckling and “Y” indicates material yielding. The complete static stress–strain curve obtained from specimen SC4L210 is shown in Fig. 4.

Initial overall geometric imperfections of the column specimens were measured prior to testing. Minor axis flexural imperfections

Table 1
Measured specimen dimensions for series C1.

Specimen	Depth	Width	Thickness	Outer radius	Inner radius	Area
	<i>D</i> (mm)	<i>B</i> (mm)	<i>t</i> (mm)	<i>r_o</i> (mm)	<i>r_i</i> (mm)	<i>A</i> (mm ²)
SC1L150 ^a	50.2	30.3	2.557	5.0	1.8	364.3
C1L200	50.0	30.0	2.571	5.0	1.8	364.2
C1L550	50.1	30.1	2.563	4.7	1.7	363.2
C1L900	50.5	30.2	2.574	4.7	1.8	359.7
C1L1200	50.1	30.2	2.560	4.8	1.7	360.7
C1L1550	50.4	30.1	2.551	5.0	2.0	364.4
Mean	50.2	30.1	2.563	4.9	1.8	362.4
COV	0.004	0.003	0.003	0.031	0.070	0.005

^a Reported in Huang and Young [4].

Table 3
Measured specimen dimensions for series C3.

Specimen	Depth	Width	Thickness	Outer radius	Inner radius	Area
	<i>D</i> (mm)	<i>B</i> (mm)	<i>t</i> (mm)	<i>r_o</i> (mm)	<i>r_i</i> (mm)	<i>A</i> (mm ²)
SC3L150 ^a	50.3	49.7	2.487	3.2	1.0	458.6
C3L200	50.1	49.6	2.492	3.0	1.2	457.4
C3L550	50.3	49.8	2.493	2.8	1.0	458.7
C3L900	50.1	50.0	2.487	3.0	1.2	457.4
C3L1200	50.2	50.0	2.520	3.2	1.0	466.6
C3L1550	50.3	50.0	2.489	2.8	1.0	460.1
Mean	50.2	49.9	2.496	3.0	1.1	460.0
COV	0.006	0.004	0.005	0.052	0.082	0.008

^a Reported in Huang and Young [4].

Table 4
Measured specimen dimensions for series C4.

Specimen	Depth	Width	Thickness	Outer radius	Inner radius	Area
	<i>D</i> (mm)	<i>B</i> (mm)	<i>t</i> (mm)	<i>r_o</i> (mm)	<i>r_i</i> (mm)	<i>A</i> (mm ²)
SC4L210 ^a	70.4	50.8	2.523	4.0	1.0	570.3
C4L200	70.4	50.5	2.537	3.8	1.2	570.5
C4L550	70.7	51.0	2.517	4.2	1.2	570.5
C4L900	70.5	50.7	2.553	4.2	1.0	575.1
C4L1200	70.4	50.7	2.513	4.0	1.0	565.7
C4L1550	70.5	50.8	2.527	3.8	1.2	571.0
Mean	70.5	50.7	2.529	4.0	1.1	570.6
COV	0.001	0.004	0.006	0.037	0.084	0.006

^a Reported in Huang and Young [4].

were recorded for all specimens, except for the stub columns. Theodolite was used to obtain readings at mid-length and near both ends of the specimens. The overall geometric imperfections at mid-length (δ) normalized with respect to the specimen length (L) are summarized in Table 9. The average overall geometric imperfections at mid-length were 1/2085, 1/2167, 1/2157, 1/2435, 1/1992 and 1/2155 of the specimen length for series C1, C2, C3, C4, C5 and C6, respectively. The maximum initial local geometric imperfections of the specimens are 0.066, 0.310, 0.101, 0.226, 0.348 and 0.679 mm for series C1, C2, C3, C4, C5 and C6, respectively, as detailed in Huang and Young [4].

4. Column tests

4.1. Test setup and test procedure

The lean duplex stainless steel columns were compressed between pinned ends. A hydraulic testing machine was used to apply compressive force to the specimens. The test rig and the test setup of a typical column are shown in Figs. 5 and 6. The pin-ended bearings were used at both upper and lower end supports to allow free rotation of the specimens about the minor axis only. Each pin-ended bearing is made up of the wedge plate and the pit plate. The wedge plate has slot holes to allow adjustment of the column specimens to be loaded concentrically. One of the pit plates was connected to a rigid plate at the upper end support, while the other pit plate was connected to the special bearing at the lower end support. The end plates were welded to the ends of the column specimens, and bolted to the two wedge plates. The specimens were then put into the testing machine between the two pit plates, so that the knife-edge of the bottom wedge plate was seated on the V-shape pit of the bottom pit plate. Initially, the special bearing was free to rotate in any direction. The bottom and upper wedges were positioned in-line and then applied approximately 5 kN to the specimens. This procedure would eliminate any possible gap between the wedge plates and the pit plates in the

Table 5
Measured specimen dimensions for series C5.

Specimen	Depth	Width	Thickness	Outer radius	Inner radius	Area
	<i>D</i> (mm)	<i>B</i> (mm)	<i>t</i> (mm)	<i>r_o</i> (mm)	<i>r_i</i> (mm)	<i>A</i> (mm ²)
SC5L300 ^a	100.1	50.9	2.510	3.3	1.2	718.0
C5L200	100.1	50.1	2.500	3.7	1.2	710.8
C5L550	100.1	50.5	2.487	3.3	1.0	709.3
C5L900	100.4	50.4	2.497	3.3	1.2	713.9
C5L900R	100.1	50.4	2.490	3.5	1.0	709.7
C5L1200	100.1	50.5	2.480	3.5	1.0	708.3
C5L1200R	100.6	50.8	2.513	3.7	1.0	720.3
C5L1550	100.2	50.4	2.510	3.5	1.2	716.8
Mean	100.2	50.4	2.497	3.5	1.1	712.7
COV	0.002	0.004	0.005	0.040	0.083	0.006

^a Reported in Huang and Young [4].**Table 6**
Measured specimen dimensions for series C6.

Specimen	Depth	Width	Thickness	Outer radius	Inner radius	Area
	<i>D</i> (mm)	<i>B</i> (mm)	<i>t</i> (mm)	<i>r_o</i> (mm)	<i>r_i</i> (mm)	<i>A</i> (mm ²)
SC6L450 ^a	150.0	50.2	2.463	4.3	2.0	944.8
C6L200	150.0	50.1	2.501	4.3	2.2	957.7
C6L550	150.1	50.2	2.490	4.5	2.0	953.0
C6L900	150.4	50.3	2.473	4.7	2.2	950.0
C6L1200	149.9	50.5	2.459	4.5	1.8	944.7
C6L1550	150.5	50.3	2.487	4.5	2.0	956.8
Mean	150.2	50.3	2.477	4.5	2.0	952.4
COV	0.002	0.003	0.006	0.030	0.062	0.006

^a Reported in Huang and Young [4].

pin-ended bearings, since the special bearing was free to rotate in any direction. The special bearing was then restrained from twisting and rotation by using the horizontal and vertical bolts, respectively. The load on the specimens was then released to approximately 2 kN to ensure the pin-ended bearings are in full contact and the test starts with a small initial load.

Three displacement transducers (LVDTs) were used to measure the axial shortening and end rotation of the specimens. In addition, two displacement transducers were located horizontally at mid-length of the specimens to measure the horizontal deflection of the specimens about the bending axis. Strain gauges were attached in the axial direction at mid-length of the specimens to determine the loading eccentricity and local buckling. Four strain gauges were located on the webs near the corners for all specimens. Two additional strain gauges were located in the middle of the webs for columns having specimen length of 550 mm to observe the occurrence of local buckling. Displacement control was used to drive the hydraulic actuator at a constant speed of 0.2 mm/min. The static load was recorded by pausing the applied straining for 2 min near the ultimate load and post-ultimate load. A data acquisition system was used to record the applied load, the readings of the LVDTs and strain gauges at regular intervals during the tests.

4.2. Concentrically loading

The pin-ended columns were aiming at zero eccentricity. Therefore, four strain gauges were used to determine the loading eccentricity (e). The applied load, longitudinal strains and overall deflection at mid-length about the bending axis of the specimens were recorded to determine the loading eccentricity. The bending moment of the columns at mid-length equals to $E_o I_y \kappa$, and also equals to the axial force (P) multiply by the summary of eccentricity, initial overall imperfection and lateral deflection. Therefore, the measured loading eccentricity including the initial overall geometric imperfection (δ) is determined by $(e + \delta) = (E_o I_y \kappa / P) - u$, where E_o = initial Young's modulus; I_y = second moment area about the minor axis; κ = curvature of the specimens which was calculated using the readings of the strain gauges; P = applied compressive load; and u = overall horizontal deflection at mid-length about the bending axis of the specimens. The measured loading eccentricity was calculated for each specimen during the initial stage of loading in the elastic range. When the eccentricity was not near zero, then the specimens were adjusted until an acceptable eccentricity was achieved. By applying this method, the largest eccentricity ($e + \delta$) of the specimens is 0.67 mm for specimen C5L1200, and the average eccentricity for all the specimens is 0.33 mm. The values of the eccentricity for each column are shown in Tables 10–15.

4.3. Test results

The experimental ultimate loads (P_{Exp}) and failure modes of the pin-ended columns are shown in Tables 10–15 for series C1, C2, C3, C4, C5 and C6, respectively. The effective length (l_e) of the column

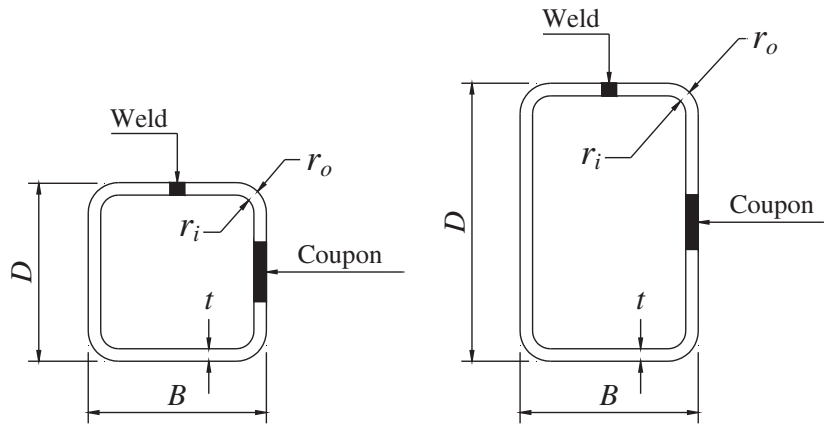


Fig. 1. Definition of symbols and location of tensile coupon in cross-section.

was assumed to be equal to the length between the two ends of the wedges, which is the sum of specimen length (L), the thicknesses of the two end plates (40 mm) and the heights of the two wedge plates (70 mm). The effective lengths of the columns are also shown in Tables 10–15. Generally speaking, the stub column failure loads are greater than those pin-ended column failure loads as shown in Tables 10–15. The failure modes observed at ultimate load of the specimens involved material yielding (Y), local buckling (L), flexural buckling (F) and interaction of local and overall flexural buckling (L + F). It should be noted that the yield strength ($A\sigma_{0.2}$) of the column specimen is the 0.2% proof stress (yield stress) at flat portions multiply by the area of flat portions plus the 0.2% proof stress at the corners multiply by the area of the corners. In this study, the material yielding is defined as the ultimate strength is greater than the yield strength of the column specimen. Fig. 7 shows the local buckling failure of specimen C6L550. The flexural buckling failure of specimen C3L1550 is shown in Fig. 8. The interaction of local and flexural buckling failure of specimen C6L1200 is shown in Fig. 9.

Flexural buckling was observed for all specimens in series C1, except for the short specimens SC1L150 and C1L200 that failed by yielding of material. Flexural buckling was also observed for specimens of effective lengths greater than or equal to 1010 mm for series C2–C5, except for specimens C5L900 and C5L900R that failed by interaction of local and flexural buckling. Local buckling was observed for specimens having effective lengths less than or equal to 660 mm for series C2–C6, except for the short specimens SC3L150, C3L200, SC4L210 and C4L200 that failed by material yielding. Interaction of local and flexural buckling was observed for specimens C6L900, C6L1200 and C6L1550. Fig. 10 shows the load-axial shortening curve of specimen C3L900. Two pin-ended column tests were repeated and the test results for the repeated tests are very close to their first test values, with differences ranging from 0.2% to 1.3%. The small differences between the repeated test values and their first test values demonstrated the reliability of the test results.

Table 7
Measured material properties obtained from tensile coupon tests [4].

Section ($D \times B \times t$)	$\sigma_{0.2}$ (MPa)	σ_u (MPa)	E_s (GPa)	ε_f (%)	n
50 × 30 × 2.5	683	764	198	39	7
50 × 50 × 1.5	610	734	194	44	5
50 × 50 × 2.5	635	756	198	44	6
70 × 50 × 2.5	613	738	194	44	8
100 × 50 × 2.5	625	727	200	49	6
150 × 50 × 2.5	664	788	202	35	4

5. Reliability analysis

Reliability analysis serves as an important tool in this study to assess the reliability of different design rules to lean duplex stainless steel columns, which is not covered in the current design specifications. Reliability analysis is detailed in the Commentary of the ASCE Specifications [11]. A target reliability index (β_0) of 2.5 for stainless steel structural members is used as a lower limit. The design rules are considered to be reliable if the reliability index is greater than 2.5. The resistance factors (ϕ_0) of 0.85, 0.90 and 0.91 for concentrically loaded compression members, and the load combinations of 1.2DL + 1.6LL, 1.25DL + 1.5LL and 1.35DL + 1.5LL as recommended by ASCE [11], AS/NZS [12] and EC3 [13] specifications, respectively, were used in the reliability analysis, where DL is the dead load and LL is the live load. Eq. 6.2-2 in the ASCE Specification was used in calculating the reliability index. The statistical parameters $M_m = 1.10$, $F_m = 1.00$, $V_m = 0.10$ and $V_F = 0.05$, which are the mean values and coefficients of variation for material properties and fabrication factors for flexural buckling members in the commentary of the ASCE Specification were adopted. In calculating the reliability index, the correction factor Eq. F1.1-3 in the North American cold-formed steel Specification AISI S100 [14] was used to account for the influence due to a small number of tests. For the purpose of direct comparison, a constant resistance factor (ϕ_1) of 0.85 and a load combination of 1.2DL + 1.6LL as specified in the ASCE Specifications were used to calculate the reliability index (β_1) for the AS/NZS and EC3 Specifications. The values of the reliability index are shown in Table 16 for all columns. Series C1–C6 of the pin-ended column specimens in this study are separately presented in Tables 10–15.

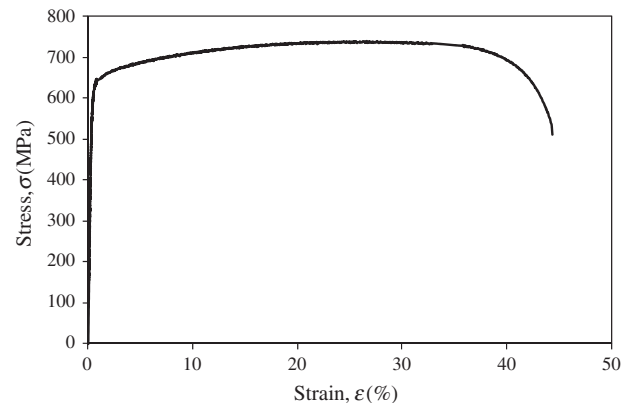


Fig. 2. Complete static stress–strain curve obtained from tensile coupon test for series C4.

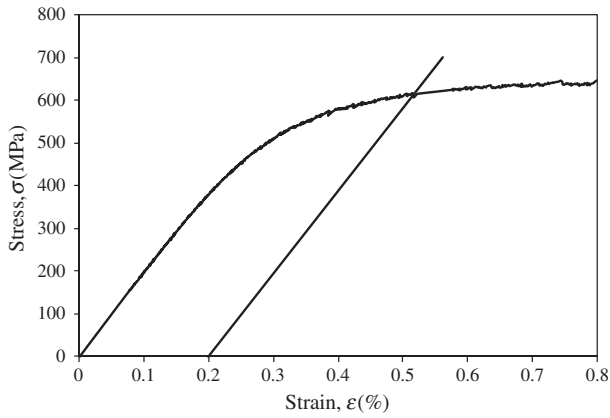


Fig. 3. Initial part of static stress–strain curve obtained from tensile coupon test for series C4.

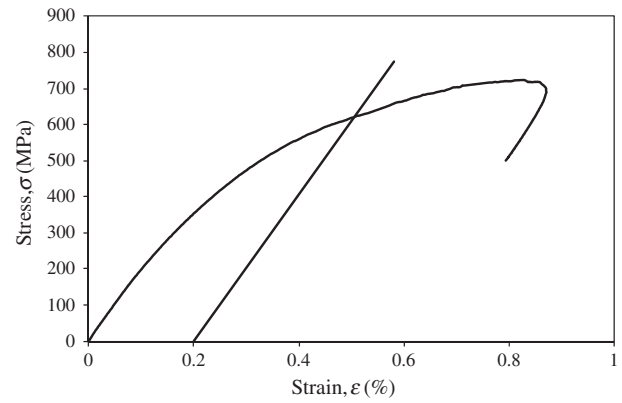


Fig. 4. Complete static stress–strain curve obtained from stub column specimen SC4L210.

6. Current design rules and comparison with test strengths

6.1. General

The applicability of current design rules to lean duplex stainless steel columns is assessed by comparing the design strengths with the test strengths that obtained from this study and previous researches [4,5]. The unfactored design strengths (nominal strength) P_{ASCE} , $P_{AS/NZS}$ and P_{EC3} were calculated using the material properties obtained from the tensile coupon tests, whereas P_{ASCE}^* , $P_{AS/NZS}^*$ and P_{EC3}^* were calculated using the material properties obtained from the stub column tests for ASCE Specification [11], AS/NZS Standard [12], and EC3 Code [13], respectively. It should be noted that the lean duplex stainless steel is not covered in these three design specifications. The design strengths (P_{ASCE} , $P_{AS/NZS}$, and P_{EC3}) calculated based on coupon test results are compared with 58 test strengths, including 32 pin-ended column strengths in this study, 6 stub column strengths in Huang and Young [4], 12 pin-ended column strengths and 8 stub column strengths in Theofanous and Gardner [5]. The material properties obtained from the stub column tests were not reported in Theofanous and Gardner [5]. Therefore, the design strengths (P_{ASCE}^* , $P_{AS/NZS}^*$, and P_{EC3}^*) calculated based on stub column test results are compared with the pin-ended column test strengths obtained in this study and the stub column test strengths reported by Huang and Young [4]. The comparison of test strengths with the design strengths for all column specimens is shown in Table 16. The series C1–C6 of the test specimens in this study is separately presented in Tables 10–15 and Figs. 11–16, respectively.

The effective length (l_e) of the column was assumed to be equal to the length between the two ends of the wedges. The effective length factor (k) of pin-ended column was taken as 1.0. Therefore, the effective length of the pin-ended column is the specimen length plus the heights of the pin-ended bearings as shown in Tables 10–15. The effective length of the stub column was taken as half of the specimen length. It

is worth noting that the design strengths were calculated based on the design rules for cold-formed stainless steel concentrically loaded compression members. However, a small eccentricity (e) was found at the ends of the columns due to the setup of the pin-ended columns. When the columns subjected to axial load, the eccentricity together with the initial overall imperfection (δ) caused an extra bending moment on the columns. The measured values of $e + \delta$ for each test specimen are summarized in Tables 10–15. In order to investigate the effect of these extra bending moments on the design strengths, the column strengths calculated by considering zero eccentricity (concentrically loaded columns) are compared with the column strengths calculated using the measured values $e + \delta$. The material properties obtained from tensile coupon tests were used in the calculation. For ASCE Specification, the largest differences between the two calculations

Table 9
Measured overall geometric imperfections at mid-length of column.

Specimen	δ/L	Specimen length L (mm)
C1L200	1/1575	200
C1L550	1/1732	550
C1L900	1/2834	900
C1L1200	1/2362	1200
C1L1550	1/2441	1550
C2L200	1/1050	200
C2L550	1/2165	550
C2L900	1/2362	900
C2L1200	1/3779	1200
C2L1550	1/4882	1550
C3L200	1/1050	200
C3L550	1/2165	550
C3L900	1/2835	900
C3L1200	1/3779	1200
C3L1550	1/3487	1550
C4L200	1/1575	200
C4L550	1/2165	550
C4L900	1/2835	900
C4L1200	1/3150	1200
C4L1550	1/3487	1550
C5L200	1/1050	200
C5L550	1/2165	550
C5L900	1/2362	900
C5L900R	1/2834	900
C5L1200	1/2700	1200
C5L1200R	1/2700	1200
C5L1550	1/2219	1550
C6L200	1/1050	200
C6L550	1/2887	550
C6L900	1/2835	900
C6L1200	1/2362	1200
C6L1550	1/4068	1550

Table 8
Measured material properties obtained from stub column tests [4].

Specimen	Section ($D \times B \times t$)	$\sigma_{0.2}$ (MPa)	σ_u (MPa)	E_o (GPa)	n	Failure mode
SC1L150	50 × 30 × 2.5	660	880	205	4	Y
SC2L150	50 × 50 × 1.5	N/A	515	194	4	L
SC3L150	50 × 50 × 2.5	690	791	202	3	Y
SC4L210	70 × 50 × 2.5	621	723	204	3	Y
SC5L300	100 × 50 × 2.5	545	548	207	4	L
SC6L450	150 × 50 × 2.5	430	452	202	8	L

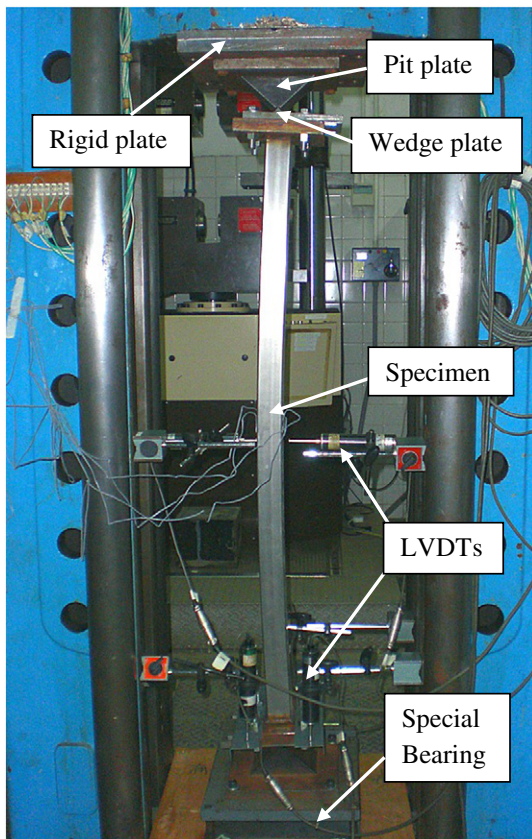


Fig. 5. Pin-ended column test of specimen C3L900.

for series C1 to C6 are 3.0%, 3.3%, 2.7%, 3.3%, 3.5% and 2.8%, respectively. For AS/NZS Standard, the largest differences are 2.9%, 3.4%, 2.8%, 2.8%, 2.9% and 2.1% for series C1 to C6, respectively. For EC3 Code, the largest differences are 2.6%, 2.8%, 2.6%, 3.2%, 3.2% and 2.2% for series C1 to C6, respectively. It is found that the effect between these two calculations is not larger than 3.5%.

It should be also noted that the 0.2% proof stress ($\sigma_{0.2}$) of stub column specimen SC2L150 could not be reached due to local buckling. According to clause 6.3 of the ASCE Specification [11], the compressive

yield stresses shall be taken as either the maximum compressive strength of the section divided by the cross-section area or the stress determined by the 0.2% offset method, whichever is reached first in the test. Therefore, the static tensile strength (σ_u) was adopted as the yield stress in calculating P_{ASCE}^* , $P_{AS/NZS}^*$ and P_{EC3}^* for column specimens in series C2.

6.2. American specification

In the design rules of ASCE Specification, the calculation of design strengths (P_n) for the column specimens are detailed in clause 3.4 of the specification for concentrically loaded compression members. This design procedure involves iterative process, which is inconvenient compare with the other design rules. In calculating the effective area (A_e), effective widths were obtained according to clause 2.2.1 of the specification, and the corner areas were considered as fully effective. Considering the strength increase due to cold-working, the material properties shall be determined on the basis of full sections, which are the stub column tests, according to clauses 1.5.4 and 6.3 of the specification. It should be noted that such mechanical properties cannot be used if the value of ρ calculated by clause 2.2.1 is less than unity ($\rho < 1$), meaning that the section is not fully effective (slender section).

It is shown in Tables 10–15 that the design strengths (P_{ASCE}) are unconservative for series C1 and C2, except for specimens SC1L150, C1L200 and C2L900. For series C3 to C6, ASCE Specification provides generally conservative prediction except for specimens with long effective lengths (l_e) of 1310 and 1660 mm in series C3 to C5, and specimens C6L200 and C6L550. The mean values of P_{Exp}/P_{ASCE} ratio are ranging from 0.95 to 1.09, and the values of COV ranged from 0.048 to 0.150, as shown in Tables 10–15. Despite the fact that the mean values of P_{Exp}/P_{ASCE} ratio calculated by the ASCE Specification based on coupon test results are close to unity ($P_m = 1.0$), series C1 and C3 could not reach the target reliability index of 2.5. According to the specification, material properties obtained from stub column tests can only apply to slender sections. However, the specimens of effective lengths less than or equal to 660 mm for series C2 and C4, and all specimens of series C5 and C6 do not comply with such requirement and identified by the symbol “^” in Tables 10–15. It is shown in Table 16 that the ASCE Specification provides slightly conservative and reliable predictions for all column specimens. The mean value of P_{Exp}/P_{ASCE} and P_{Exp}/P_{ASCE}^* equal to 1.03 and 1.12, with COV of 0.116 and 0.112, respectively. The reliability indices (β_0 and β_1) are greater than the target value of 2.5.

6.3. Australian/New Zealand standard

In AS/NZS Standard, the column design strengths (N_c) were calculated using the alternative design method given in clause 3.4.2 of the standard. The calculation of buckling stress (f_n) requires the determination of values α , β , λ_0 and λ_1 according to the type of materials. It should be noted that the values of lean duplex material are not given in the AS/NZS Standard. Therefore, the values of S31803 (duplex) given in Table 3.4.2 of the AS/NZS Standard were used to calculate the buckling stress in this study. The calculation of effective areas was based on clause 2.2.1.2 of the standard, which used the same effective width equations as those of the ASCE Specification except for the calculation of flexural buckling stress. Similarly, the corner areas were considered as fully effective. According to clause 1.5.2.4 of the standard, the strength increase resulting from cold-forming shall be permitted by substituting the average design yield stress (f_{ya}) of the full section for f_y . In this study, the material properties of the full section were obtained from stub column tests. However, this design provision for the increase in strength is only applicable to ferritic stainless steel and fully effective section, as described in clause 1.5.2.4. Therefore, this approach is not applicable for lean duplex stainless steel nor slender section.

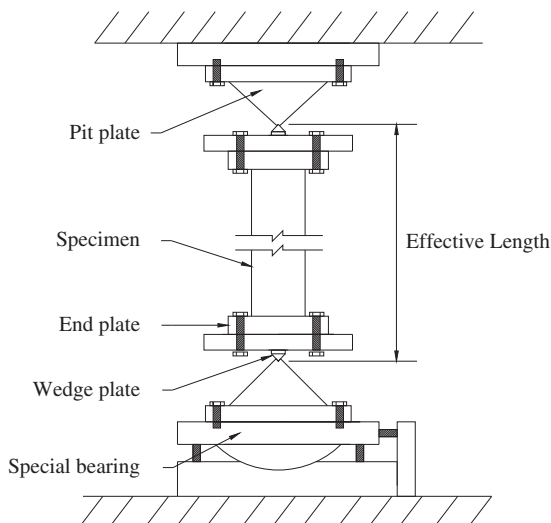


Fig. 6. Schematic view of pin-ended column setup.

Table 10
Comparison of test strengths with design strengths for series C1.

Specimen	Test				Comparison								
	P_{Exp} (kN)	Failure mode	l_e (mm)	$e + \delta$ (mm)	$\frac{P_{Exp}}{P_{ASCE}}$	$\frac{P_{Exp}}{P_{AS/NZS}}$	$\frac{P_{Exp}}{P_{EC3}}$	$\frac{P_{Exp}}{P_{ASCE}^*}$	$\frac{P_{Exp}}{P_{AS/NZS}^*}$	$\frac{P_{Exp}}{P_{EC3}^*}$	$\frac{P_{Exp}}{P_{ASCE}^{\#}}$	$\frac{P_{Exp}}{P_{AS/NZS}^{\#}}$	$\frac{P_{Exp}}{P_{EC3}^{\#}}$
SC1L150 ^a	316.1	Y	75	–	1.23	1.26	1.27	1.28	1.31	1.31	1.28	1.28	1.31
C1L200	259.2	Y	310	0.36	1.05	1.10	1.10	1.05	1.12	1.13	1.05	1.10	1.13
C1L550	154.2	F	660	0.01	0.90	1.10	1.12	1.01	1.11	1.11	1.01	1.10	1.11
C1L900	85.4	F	1010	0.27	0.86	1.06	1.15	0.93	1.05	1.13	0.92	1.03	1.13
C1L1200	55.3	F	1310	0.14	0.93	1.06	1.19	0.93	1.05	1.15	0.92	1.02	1.15
C1L1550	36.2	F	1660	0.19	0.97	1.05	1.19	0.95	1.04	1.16	0.95	1.02	1.16
Mean (P_m)					0.99	1.11	1.17	1.02	1.11	1.16	1.02	1.09	1.16
COV(V_p)					0.136	0.073	0.052	0.130	0.092	0.065	0.133	0.090	0.065
Resistance factor (ϕ_0)					0.85	0.90	0.91	0.85	0.90	0.91	0.85	0.90	0.91
Reliability index (β_0)					2.20	2.60	2.93	2.35	2.50	2.85	2.31	2.44	2.85
Resistance factor (ϕ_1)					0.85	0.85	0.85	0.85	0.85	0.85	0.85	0.85	0.85
Reliability index (β_1)					2.20	3.01	3.36	2.35	2.90	3.26	2.31	2.84	3.26

* Calculated using material properties obtained from stub column tests.

Calculated using material properties obtained from stub column tests and full cross-section area for all class of sections (including slender sections).

^a Reported in Huang and Young [4].

For AS/NZS Standard, the design strengths ($P_{AS/NZS}$) of the specimens calculated based on material properties from coupon test results are conservative, except for specimens SC2L150, C2L200, C6L200 and C6L550. The mean values of $P_{Exp}/P_{AS/NZS}$ ratio are ranging from 1.03 to 1.21, and the values of COV ranged from 0.048 to 0.101, as shown in Tables 10–15. The reliability indices are greater than the target value of 2.5 for all series, except for the reliability index (β_0) of series C2 and C6. According to the standard, the specimens of effective lengths less than or equal to 660 mm for series C2 and C4, except for specimen C4L550, and all specimens of series C5 and C6, except for specimen C5L1550, do not comply with the aforementioned requirement of material properties obtained from stub column tests as identified by the symbol “^” in Tables 10–15. It is shown in Table 16 that the AS/NZS Standard is conservative and reliable for all column specimens. The mean value of $P_{Exp}/P_{AS/NZS}$ and $P_{Exp}/P_{AS/NZS}^*$ equal to 1.14 and 1.19, with COV of 0.083 and 0.091, respectively. The reliability indices (β_0 and β_1) are greater than the target value of 2.5.

6.4. European code

In the European code, according to clause 5.4.1 of the EC3 Part 1.4, the column design strengths ($N_{b,Rd}$) were obtained using clause 6.3.1.1 of the EC3 Part 1.1: General rules and rules for buildings [15], except

that the reduction factor (χ) was calculated according to clause 5.4.2.1 of the EC3 Part 1.4 for uniform members in compression. The imperfection factor $\alpha = 0.49$ and limiting slenderness $\bar{\lambda}_0 = 0.4$ for hollow sections failed by flexural buckling as indicated in Table 5.3 of the EC3 Part 1.4 were used. Section classification of the specimens according to Table 5.2 of the EC3 Part 1.4 for maximum width-to-thickness ratios for compression parts are required to calculate the design strengths. In this study, the sections of series C1 and C3 were classified as class 3 cross-sections, whereas the sections of series C2, C4, C5 and C6 were classified as class 4 cross-sections. Therefore, effective areas (A_e) were used instead of full cross-section area (A) in calculating the design strengths of class 4 cross-sections. According to clause 5.2.3 of the EC3 Part 1.4, the effective widths (b_{eff}) were calculated based on EC3 Part 1.5: Plated structural elements [16], except that the reduction factor (ρ) was calculated by Eq. 5.1 of the EC3 Part 1.4 for cold formed or welded internal elements. The corner areas were assumed as fully effective. In clause 2.1 of the EC3 Part 1.4, it is reference to Annex A clause 3.2.1 of the EC3 Part 1.3: General rules – Supplementary rules for cold-formed members and sheeting [17] that the material properties may be obtained from stub column tests, which is applicable to fully effective section only, as indicated in clause 3.2.2 of the EC3 Part 1.3. It should be noted that for stub columns failed by local buckling, the material properties have already included the effect of local buckling.

Table 11
Comparison of test strengths with design strengths for series C2.

Specimen	Test				Comparison								
	P_{Exp} (kN)	Failure mode	l_e (mm)	$e + \delta$ (mm)	$\frac{P_{Exp}}{P_{ASCE}}$	$\frac{P_{Exp}}{P_{AS/NZS}}$	$\frac{P_{Exp}}{P_{EC3}}$	$\frac{P_{Exp}}{P_{ASCE}^*}$	$\frac{P_{Exp}}{P_{AS/NZS}^*}$	$\frac{P_{Exp}}{P_{EC3}^*}$	$\frac{P_{Exp}}{P_{ASCE}^{\#}}$	$\frac{P_{Exp}}{P_{AS/NZS}^{\#}}$	$\frac{P_{Exp}}{P_{EC3}^{\#}}$
SC2L150 ^a	147.1	L	75	–	0.95	0.95	1.12	1.07 [^]	1.08 [^]	1.25 [^]	1.06	1.06	1.06
C2L200	153.4	L	310	0.39	0.97	0.97	1.13	1.09 [^]	1.15 [^]	1.26 [^]	1.08	1.08	1.08
C2L550	139.3	L	660	0.48	0.97	1.05	1.14	1.06 [^]	1.14 [^]	1.25 [^]	1.02	1.13	1.06
C2L900	120.8	F	1010	0.51	1.01	1.10	1.23	1.14	1.23	1.32 [^]	1.13	1.21	1.13
C2L1200	92.3	F	1310	0.41	0.94	1.09	1.23	1.10	1.16	1.29 [^]	1.09	1.16	1.15
C2L1550	65.4	F	1660	0.48	0.87	1.03	1.17	0.99	1.12	1.21 [^]	0.99	1.08	1.13
Mean (P_m)					0.95	1.03	1.17	1.08	1.15	1.26	1.06	1.12	1.10
COV(V_p)					0.048	0.060	0.043	0.046	0.043	0.030	0.047	0.051	0.035
Resistance factor (ϕ_0)					0.85	0.90	0.91	0.85	0.90	0.91	0.85	0.90	0.91
Reliability index (β_0)					2.54	2.39	2.97	3.05	2.88	3.34	2.99	2.76	2.75
Resistance factor (ϕ_1)					0.85	0.85	0.85	0.85	0.85	0.85	0.85	0.85	0.85
Reliability index (β_1)					2.54	2.82	3.40	3.05	3.32	3.77	2.99	3.19	3.18

* Calculated using material properties obtained from stub column tests.

Calculated using material properties obtained from stub column tests and full cross-section area for all class of sections (including slender sections).

^a Reported in Huang and Young [4].

[^] Specifications do not allow material properties determine from stub column tests for slender sections ($\rho < 1$).

Table 12
Comparison of test strengths with design strengths for series C3.

Specimen	Test				Comparison								
	P_{Exp} (kN)	Failure mode	l_e (mm)	$e + \delta$ (mm)	$\frac{P_{Exp}}{P_{ASCE}}$	$\frac{P_{Exp}}{P_{AS/NZS}}$	$\frac{P_{Exp}}{P_{EC3}}$	$\frac{P_{Exp}}{P_{ASCE}^*}$	$\frac{P_{Exp}}{P_{AS/NZS}^*}$	$\frac{P_{Exp}}{P_{EC3}^*}$	$\frac{P_{Exp}}{P_{ASCE}^\#}$	$\frac{P_{Exp}}{P_{AS/NZS}^\#}$	$\frac{P_{Exp}}{P_{EC3}^\#}$
SC3L150 ^a	362.2	Y	75	–	1.23	1.23	1.24	1.13	1.13	1.14	1.13	1.13	1.14
C3L200	355.3	Y	310	0.13	1.21	1.21	1.22	1.11	1.11	1.13	1.11	1.11	1.13
C3L550	302.1	L	660	0.05	1.16	1.30	1.20	0.99	1.19	1.12	0.97	1.19	1.12
C3L900	242.8	F	1010	0.05	1.19	1.38	1.34	1.22	1.27	1.28	1.18	1.26	1.28
C3L1200	146.1	F	1310	0.37	0.88	1.07	1.10	0.99	1.05	1.06	0.99	1.02	1.06
C3L1550	103.9	F	1660	0.55	0.89	1.08	1.15	0.99	1.03	1.12	0.97	1.02	1.12
Mean (P_m)					1.09	1.21	1.21	1.07	1.13	1.14	1.06	1.12	1.14
COV(V_p)					0.150	0.101	0.068	0.091	0.080	0.063	0.087	0.084	0.063
Resistance factor (ϕ_0)					0.85	0.90	0.91	0.85	0.90	0.91	0.85	0.90	0.91
Reliability index (β_0)					2.43	2.76	2.98	2.78	2.63	2.78	2.74	2.58	2.77
Resistance factor (ϕ_1)					0.85	0.85	0.85	0.85	0.85	0.85	0.85	0.85	0.85
Reliability index (β_1)					2.43	3.15	3.39	2.78	3.04	3.20	2.74	2.99	3.19

* Calculated using material properties obtained from stub column tests.

Calculated using material properties obtained from stub column tests and full cross-section area for all class of sections (including slender sections).

^a Reported in Huang and Young [4].

For European Code, the design strengths (P_{EC3}) of the specimens calculated based on material properties from tensile coupon test results are generally quite conservative. The mean values of the test strength to design strength (P_{Exp}/P_{EC3}) ratio are ranging from 1.11 to 1.24, and the values of COV ranged from 0.040 to 0.068. It is observed that the values of COV in the European Code are the smallest among the three specifications for all series, which indicates a small variation among the test strength to design strength ratio. It should be noted that the reliability indices (β_0 and β_1) are larger than the target for all series. It is shown that the design strengths of P_{EC3}^* are the most conservative among the three specifications. According to the code, the design provision for the increase in strength due to cold-forming by using material properties from stub column tests is not applicable for all specimens in series C2, C4, C5 and C6, which are not fully effective, as identified by symbol “^” in Tables 10–15. It is shown in Table 16 that the EC3 Code provides the most conservative and reliable predictions for all column specimens. The mean value of P_{Exp}/P_{EC3} and P_{Exp}/P_{EC3}^* equal to 1.18 and 1.26, with COV of 0.070 and 0.099, respectively. The reliability indices (β_0 and β_1) are greater than the target value of 2.5.

7. Recommended design approach and comparison with test strengths

7.1. Stub column and full area approach

The material properties of the entire cross-section of the specimens, including the corners and flat portions, were measured by stub column tests, as shown in Table 8. Local buckling was observed for the stub column specimens SC2L150, SC5L300 and SC6L450. Therefore, the material properties obtained from these stub column tests include the effect of local buckling. Based on the stub column test results, the calculation of the design strengths have already taken account of the effect of local buckling for series C2, C5 and C6. However, the calculation of effective area in the current specifications also considered the effect of local buckling. Therefore, it is recommended that full cross-section area, instead of effective area, is used for the calculation of design strengths for all classes of sections including the slender sections when using the material properties obtained from stub column tests. Rasmussen and Rondal [18] adopted stub column properties obtained from non-slender sections ($\rho = 1$) in calculating column strengths, but slender

Table 13
Comparison of test strengths with design strengths for series C4.

Specimen	Test				Comparison								
	P_{Exp} (kN)	Failure mode	l_e (mm)	$e + \delta$ (mm)	$\frac{P_{Exp}}{P_{ASCE}}$	$\frac{P_{Exp}}{P_{AS/NZS}}$	$\frac{P_{Exp}}{P_{EC3}}$	$\frac{P_{Exp}}{P_{ASCE}^*}$	$\frac{P_{Exp}}{P_{AS/NZS}^*}$	$\frac{P_{Exp}}{P_{EC3}^*}$	$\frac{P_{Exp}}{P_{ASCE}^\#}$	$\frac{P_{Exp}}{P_{AS/NZS}^\#}$	$\frac{P_{Exp}}{P_{EC3}^\#}$
SC4L210 ^a	413.2	Y	105	–	1.19	1.22	1.29	1.17 [^]	1.21 [^]	1.27 [^]	1.17	1.17	1.17
C4L200	373.1	Y	310	0.60	1.07	1.07	1.16	1.06 [^]	1.08 [^]	1.14 [^]	1.05	1.05	1.05
C4L550	352.8	L	660	0.52	1.12	1.21	1.22	1.00 [^]	1.20	1.19 [^]	1.00	1.20	1.11
C4L900	270.7	F	1010	0.61	1.01	1.20	1.21	1.09	1.20	1.17 [^]	1.09	1.17	1.10
C4L1200	211.2	F	1310	0.20	0.95	1.22	1.28	1.15	1.19	1.23 [^]	1.14	1.17	1.17
C4L1550	148.3	F	1660	0.39	0.91	1.16	1.26	1.08	1.14	1.20 [^]	1.08	1.11	1.16
Mean (P_m)					1.04	1.18	1.24	1.09	1.17	1.20	1.09	1.14	1.13
COV(V_P)					0.103	0.048	0.040	0.057	0.044	0.038	0.056	0.047	0.042
Resistance factor (ϕ_0)					0.85	0.90	0.91	0.85	0.90	0.91	0.85	0.90	0.91
Reliability index (β_0)					2.59	2.98	3.20	3.05	2.96	3.09	3.05	2.86	2.82
Resistance factor (ϕ_1)					0.85	0.85	0.85	0.85	0.85	0.85	0.85	0.85	0.85
Reliability index (β_1)					2.59	3.41	3.64	3.05	3.40	3.52	3.05	3.30	3.26

* Calculated using material properties obtained from stub column tests.

Calculated using material properties obtained from stub column tests and full cross-section area for all class of sections (including slender sections).

^a Reported in Huang and Young [4].

[^] Specifications do not allow material properties determine from stub column tests for slender sections ($\rho < 1$).

Table 14

Comparison of test strengths with design strengths for series C5.

Specimen	Test				Comparison								
	P_{Exp} (kN)	Failure mode	l_e (mm)	$e + \delta$ (mm)	$\frac{P_{Exp}}{P_{ASCE}}$	$\frac{P_{Exp}}{P_{AS/NZS}}$	$\frac{P_{Exp}}{P_{EC3}}$	$\frac{P_{Exp}}{P_{ASCE}^{\#}}$	$\frac{P_{Exp}}{P_{AS/NZS}^{\#}}$	$\frac{P_{Exp}}{P_{EC3}^{\#}}$	$\frac{P_{Exp}}{P_{ASCE}^{\#}}$	$\frac{P_{Exp}}{P_{AS/NZS}^{\#}}$	$\frac{P_{Exp}}{P_{EC3}^{\#}}$
SC5L300 ^a	394.0	L	150	–	1.05	1.07	1.17	1.17 [^]	1.19 [^]	1.30 [^]	1.01	1.01	1.01
C5L200	370.1	L	310	0.41	1.00	1.00	1.12	1.11 [^]	1.11 [^]	1.23 [^]	0.96	0.96	0.96
C5L550	372.3	L	660	0.03	1.08	1.16	1.18	1.13 [^]	1.23 [^]	1.28 [^]	0.98	1.08	1.03
C5L900	335.2	L + F	1010	0.54	1.15	1.28	1.30	1.28 [^]	1.36 [^]	1.37 [^]	1.17	1.25	1.16
C5L900R	336.0	L + F	1010	0.62	1.16	1.28	1.30	1.28 [^]	1.36 [^]	1.38 [^]	1.18	1.26	1.17
C5L1200	249.0	F	1310	0.67	1.00	1.15	1.22	1.14 [^]	1.19 [^]	1.26 [^]	1.08	1.15	1.12
C5L1200R	252.2	F	1310	0.62	0.99	1.14	1.21	1.12 [^]	1.21 [^]	1.24 [^]	1.07	1.14	1.11
C5L1550	193.7	F	1660	0.18	0.95	1.12	1.27	1.07 [^]	1.17	1.28 [^]	1.07	1.15	1.18
Mean (P_m)					1.05	1.15	1.23	1.16	1.23	1.29	1.07	1.14	1.10
COV(V_p)					0.079	0.083	0.055	0.073	0.077	0.046	0.079	0.091	0.075
Resistance factor (ϕ_0)					0.85	0.90	0.91	0.85	0.90	0.91	0.85	0.90	0.91
Reliability index (β_0)					2.80	2.69	3.10	3.24	2.99	3.38	2.84	2.52	2.53
Resistance factor (ϕ_1)					0.85	0.85	0.85	0.85	0.85	0.85	0.85	0.85	0.85
Reliability index (β_1)					2.80	3.10	3.52	3.24	3.41	3.81	2.84	2.92	2.93

*Calculated using material properties obtained from stub column tests.

#Calculated using material properties obtained from stub column tests and full cross-section area for all class of sections (including slender sections).

^a Reported in Huang and Young [4].[^] Specifications do not allow material properties determine from stub column tests for slender sections ($\rho < 1$).

sections with occurrence of local buckling are not covered. It should be noted that the current design specifications, as shown in Section 6 of this paper, do not allow the use of material properties obtained from stub column tests for slender sections. In contrast with the current design rules, the recommended stub column and full area approach fills the glaring hole of applying stub column properties to slender sections ($\rho < 1$). In this study, the design strengths $P_{ASCE}^{\#}$, $P_{AS/NZS}^{\#}$ and $P_{EC3}^{\#}$ were calculated using the recommended design approach for ASCE Specification [11], AS/NZS Standard [12] and EC3 Code [13], respectively. In the calculation, full cross-section area (A) of the test specimens as shown in Tables 1–6, instead of the effective area (A_e), was used together with the stub column properties as shown in Table 8 for all specimens.

The recommended design approach is relatively simpler compared to the current design rules, since the calculation of effective area is not required by simply using full cross-section area regardless of slender or non-slender sections. Furthermore, the classification of cross-section can be circumvented as required by the EC3 Code, and Eqs. (13) and (17) can be used for all classes of cross-sections. The comparison of the column test strengths (P_{Exp}) with the design strengths using the recommended design approach for the three specifications is shown in Tables 10–15 and Figs. 11–16.

7.2. Comparison of test strengths with design strengths

The material properties obtained from the stub column tests were not reported in Theofanous and Gardner [5]. Therefore, the design strengths ($P_{ASCE}^{\#}$, $P_{AS/NZS}^{\#}$, and $P_{EC3}^{\#}$) calculated based on the recommended design approach are compared with the test strengths obtained in this study and Huang and Young [4]. For ASCE Specification, the design strengths ($P_{ASCE}^{\#}$) calculated using the recommended design approach are generally capable of predicting the test strengths. The mean values of $P_{Exp}/P_{ASCE}^{\#}$ ratio are ranging from 0.97 to 1.09, and the values of COV ranged from 0.047 to 0.133, as shown in Tables 10–15. The reliability indices (β_0 and β_1) are larger than the target value of 2.5, except for series C1. Generally speaking, $P_{ASCE}^{\#}$ exhibit a similar accuracy as P_{ASCE} , but a smaller variation among the test strength to design strength.

For AS/NZS Standard, the design strengths ($P_{AS/NZS}^{\#}$) of the specimens are conservative, except for specimens C5L200, C6L200 and C6L550. The mean values of $P_{Exp}/P_{AS/NZS}^{\#}$ ratio are ranging from 1.03 to 1.14, and the values of COV ranged from 0.047 to 0.094, as shown in Tables 10–15. The reliability indices (β_0 and β_1) are larger than the target value of 2.5, except for reliability index (β_0) in series C1

Table 15

Comparison of test strengths with design strengths for series C6.

Specimen	Test				Comparison								
	P_{Exp} (kN)	Failure mode	l_e (mm)	$e + \delta$ (mm)	$\frac{P_{Exp}}{P_{ASCE}}$	$\frac{P_{Exp}}{P_{AS/NZS}}$	$\frac{P_{Exp}}{P_{EC3}}$	$\frac{P_{Exp}}{P_{ASCE}^{\#}}$	$\frac{P_{Exp}}{P_{AS/NZS}^{\#}}$	$\frac{P_{Exp}}{P_{EC3}^{\#}}$	$\frac{P_{Exp}}{P_{ASCE}^{\#}}$	$\frac{P_{Exp}}{P_{AS/NZS}^{\#}}$	$\frac{P_{Exp}}{P_{EC3}^{\#}}$
SC6L450 ^a	428.1	L	225	–	1.06	1.04	1.20	1.47 [^]	1.45 [^]	1.67 [^]	1.05	1.05	1.05
C6L200	404.1	L	310	0.36	0.99	0.99	1.11	1.36 [^]	1.34 [^]	1.55 [^]	0.98	0.98	0.98
C6L550	353.2	L	660	0.50	0.88	0.98	0.99	1.27 [^]	1.18 [^]	1.36 [^]	0.93	0.86	0.89
C6L900	333.5	L + F	1010	0.16	1.05	1.14	1.10	1.34 [^]	1.40 [^]	1.42 [^]	1.01	1.09	1.00
C6L1200	284.5	L + F	1310	0.02	1.09	1.17	1.13	1.27 [^]	1.36 [^]	1.40 [^]	0.98	1.12	1.05
C6L1550	230.0	L + F	1660	0.14	1.05	1.15	1.15	1.11 [^]	1.28 [^]	1.33 [^]	0.89	1.10	1.10
Mean (P_m)					1.02	1.08	1.11	1.30	1.33	1.45	0.97	1.03	1.01
COV(V_p)					0.076	0.078	0.064	0.094	0.071	0.090	0.058	0.094	0.074
Resistance factor (ϕ_0)					0.85	0.90	0.91	0.85	0.90	0.91	0.85	0.90	0.91
Reliability index (β_0)					2.67	2.47	2.68	3.47	3.33	3.52	2.59	2.22	2.25
Resistance factor (ϕ_1)					0.85	0.85	0.85	0.85	0.85	0.85	0.85	0.85	0.85
Reliability index (β_1)					2.67	2.88	3.09	3.47	3.75	3.91	2.59	2.62	2.66

*Calculated using material properties obtained from stub column tests.

#Calculated using material properties obtained from stub column tests and full cross-section area for all class of sections (including slender sections).

^a Reported in Huang and Young [4].[^] Specifications do not allow material properties determine from stub column tests for slender sections ($\rho < 1$).



Fig. 7. Local buckling of specimen C6L550.

and C6. The values of $P_{AS/NZS}^{\#}$ are slightly more accurate than the values of $P_{AS/NZS}$ in predicting the test strengths, except for series C2.

For European Code, the design strengths ($P_{EC3}^{\#}$) of the specimens are conservative, except for specimens C5L200, C6L200 and C6L550.

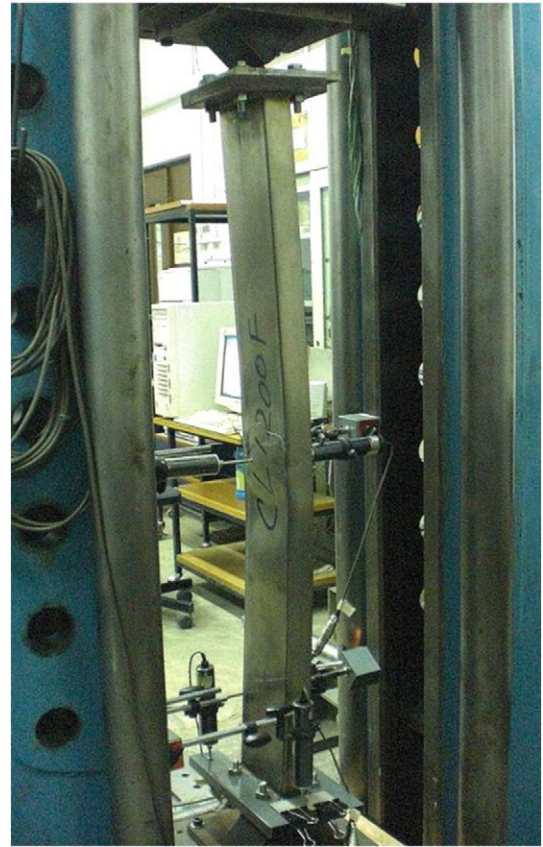


Fig. 9. Interaction of local and flexural buckling of specimen C6L1200.

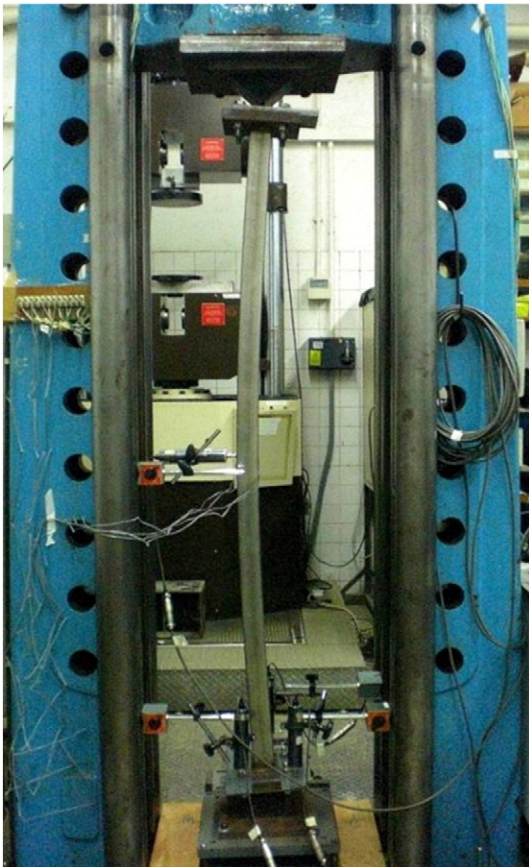


Fig. 8. Flexural buckling of specimen C3L1550.

The mean values of the test strength to design strength ($P_{Exp}/P_{EC3}^{\#}$) ratio are ranging from 1.01 to 1.16, and the values of COV ranged from 0.035 to 0.075. The reliability indices (β_0 and β_1) are generally greater than the target value, except the reliability index (β_0) based on the resistance factor (ϕ_0) of 0.91 for series C6. The design strengths of $P_{EC3}^{\#}$ are less conservative than the design strengths of P_{EC3} and P_{EC3}^* for all test series.

It is observed from the comparison of test strengths with design strengths for all test specimens that the recommended design approach provides a generally more accurate and less scattered predictions than the two design approaches using the current design rules, as shown in Table 16. In general, the design strengths calculated from

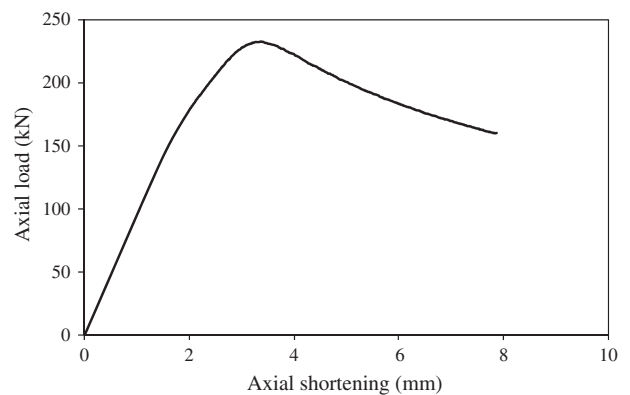


Fig. 10. Load-axial shortening curve of specimen C3L900.

Table 16
Comparison of test strengths with design strengths for all column specimens.

	$\frac{P_{Exp}}{P_{ASCE}}$	$\frac{P_{Exp}}{P_{AS/NZS}}$	$\frac{P_{Exp}}{P_{EC3}}$	$\frac{P_{Exp}}{P_{ASCE}^*}$	$\frac{P_{Exp}}{P_{AS/NZS}^*}$	$\frac{P_{Exp}}{P_{EC3}^*}$	$\frac{P_{Exp}}{P_{ASCE}^\#}$	$\frac{P_{Exp}}{P_{AS/NZS}^\#}$	$\frac{P_{Exp}}{P_{EC3}^\#}$
Number of data	58	58	58	38	38	38	38	38	38
Mean (P_m)	1.03	1.14	1.18	1.12	1.19	1.26	1.04	1.10	1.10
COV(V_P)	0.116	0.083	0.070	0.112	0.091	0.099	0.083	0.081	0.073
Resistance factor (ϕ_0)	0.85	0.90	0.91	0.85	0.90	0.91	0.85	0.90	0.91
Reliability index (β_0)	2.65	2.77	2.97	3.00	2.90	3.08	2.85	2.66	2.69
Resistance factor (ϕ_1)	0.85	0.85	0.85	0.85	0.85	0.85	0.85	0.85	0.85
Reliability index (β_1)	2.65	3.19	3.39	3.00	3.32	3.49	2.85	3.09	3.12

*Calculated using material properties obtained from stub column tests.

#Calculated using material properties obtained from stub column tests and full cross-section area for all classes of sections (including slender sections).

the design approach of using full cross-section area and material properties obtained by stub column tests, including slender sections, compared well with the test strengths. This recommended design approach does not require the calculation of effective area by using full area. Overall, the comparison shows a positive result of using the recommended design approach for the cold-formed lean duplex stainless steel columns.

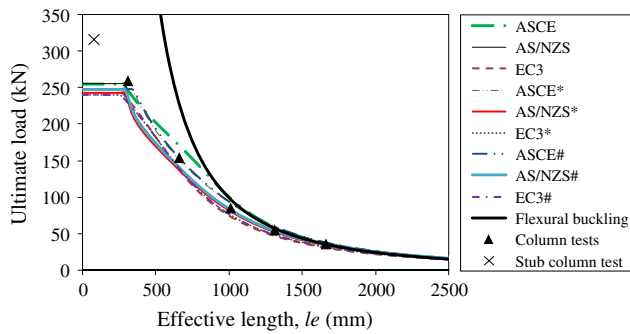


Fig. 11. Column curves for series C1.

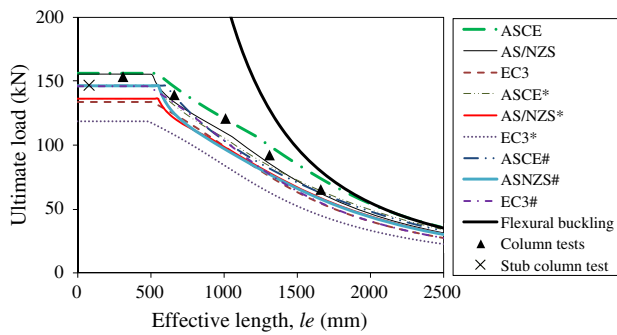


Fig. 12. Column curves for series C2.

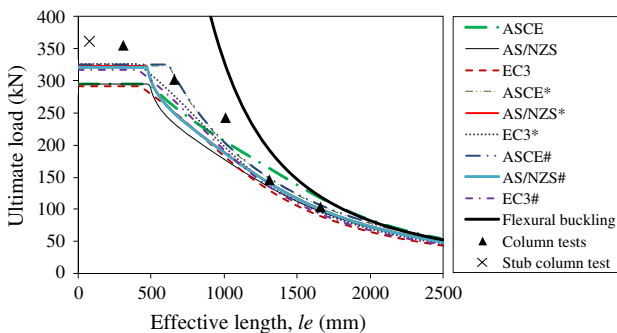


Fig. 13. Column curves for series C3.

8. Conclusions

A test program on cold-formed lean duplex stainless steel columns has been presented. The test specimens had the static yield stress and tensile strength up to 683 and 788 MPa measured from tensile coupon tests, respectively. Overall geometric imperfections of the test specimens were measured. Two series of square hollow sections and four series of rectangular hollow sections with five different column lengths for each series were concentrically compressed between pinned ends. The column curves were plotted for each test series. The test strengths were compared with the design strengths predicted using the American Specification, Australian/New Zealand Standard and European Code for stainless steel structures. It should be noted that these specifications do not cover the material of lean duplex stainless steel.

The current design specifications are generally capable of predicting the column test strengths. The design strengths calculated using the material properties from tensile coupon tests are generally quite conservative for European Code, while the American Specification predictions are less conservative. In this study, design approach by using full cross-section area and material properties obtained from stub column

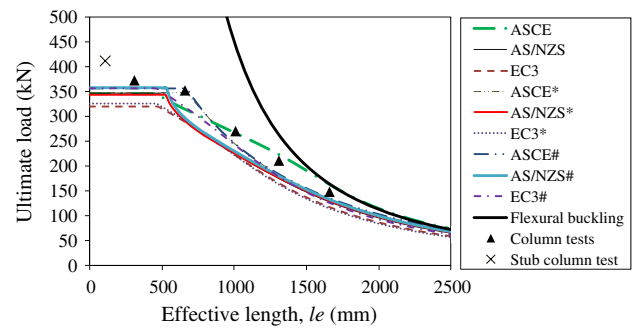


Fig. 14. Column curves for series C4.

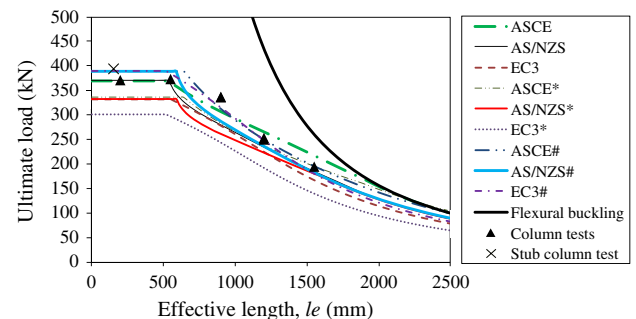


Fig. 15. Column curves for series C5.

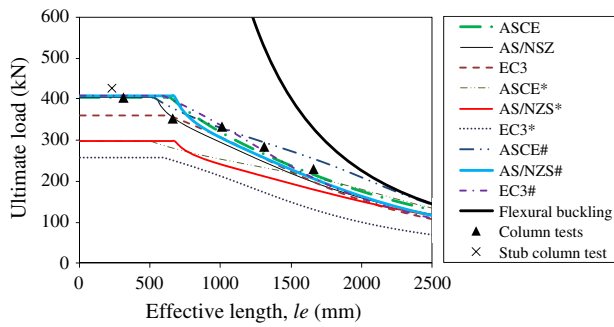


Fig. 16. Column curves for series C6.

test for both slender and non-slender sections is recommended. The recommended design approach is relatively simpler compared to the current design rules, since the calculation of effective area is not required by simply using full cross-section area. It is shown that the design strengths calculated using the recommended design approach are generally more accurate than those using the current design rules for lean duplex stainless steel columns.

Notation

The following symbols are used in this paper

A	full area;
A_e	effective area;
B	overall width of specimen;
b_{eff}	effective width in European Code;
COV	coefficient of variation;
D	overall depth of specimen;
E_o	initial Young's modulus;
e	eccentricity;
F_m	mean value of fabrication factor;
f_n	buckling stress in Australian/New Zealand Standard;
f_{ya}	average design yield stress in Australian/New Zealand Standard;
f_y	yield stress(0.2% proof stress) in Australian/New Zealand Standard and European Code;
I_y	second moment area about the minor axis;
L	length of specimen or unbraced length of member in American Specification;
l_e	effective length of specimen;
M_m	mean value of material factor;
$N_{b,Rd}$	design strength in European Code;
N_c	design strength in Australian/New Zealand Standard;
n	Ramberg–Osgood parameter;
P	axial force;
P_{ASCE}	unfactored design strengths (nominal strength) calculated using material properties obtained from tensile coupon tests for American Specification;
$P_{AS/NZS}$	unfactored design strengths calculated using material properties obtained from tensile coupon tests for Australian/New Zealand Standard;
P_{EC3}	unfactored design strengths calculated using material properties obtained from tensile coupon tests for European Code;
P_{Exp}	experimental ultimate load (test strength);
P_m	mean value of tested-to-predicted load ratio;
P_n	design strength in American Specification;
$P_{ASCE}^{\#}$	unfactored design strengths calculated using the recommended design approach for American Specification;
$P_{AS/NZS}^{\#}$	unfactored design strengths calculated using the recommended design approach for Australian/New Zealand Standard;
$P_{EC3}^{\#}$	unfactored design strengths calculated using the recommended design approach for European Code;

$P_{ASCE}^{\#}$

unfactored design strengths calculated using material properties obtained from stub column tests for American Specification;

$P_{AS/NZS}^{\#}$

unfactored design strengths calculated using material properties obtained from stub column tests for Australian/New Zealand Standard;

$P_{EC3}^{\#}$

unfactored design strengths calculated using material properties obtained from stub column tests for European Code;

r_i

inner corner radius of specimen;

r_o

outer corner radius of specimen;

r_y

radius of gyration about minor axis;

t

thickness of specimen;

u

lateral deflection at mid-length;

V_F

coefficient of variation of fabrication factor;

V_m

coefficient of variation of material factor;

V_p

coefficient of variation of tested-to-predicted load ratio;

α

coefficient of buckling stress in Australian/New Zealand Standard or imperfection factor in European Code;

β

coefficient of buckling stress in Australian/New Zealand Standard;

β_o

reliability index;

β_1

reliability index;

χ

reduction factor in European Code;

δ

initial overall imperfection of column;

ε_f

tensile strain after fracture based on gauge length of 25 mm;

ϕ_o

resistance factor;

ϕ_1

resistance factor;

λ_o

coefficient of buckling stress in Australian/New Zealand Standard;

$\bar{\lambda}_o$

limiting slenderness in European Code;

λ

coefficient of buckling stress in Australian/New Zealand Standard;

κ

curvature;

ρ

value in calculating effective area in American Specification, Australian/New Zealand Standard and European Code;

σ_u

static tensile strength;

$\sigma_{0.01}$

static 0.01% tensile proof stress; and

$\sigma_{0.2}$

static 0.2% tensile proof stress.

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